

SpeechEditBench: A Bilingual Multi-Attribute Benchmark for Instruction-Guided Speech Editing

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Can speech editing models **reliably satisfy editing instructions** while **preserving untargeted attributes**?

Why SpeechEditBench?

Existing evaluations are **fragmented by task** and often score only whether the requested change happened.

Editing is a **dual-constraint** problem:
change the target · **preserve the rest**

1 Unified coverage

7 atomic tasks plus 2- and 3-way compositional editing.

2 Bilingual diagnosis

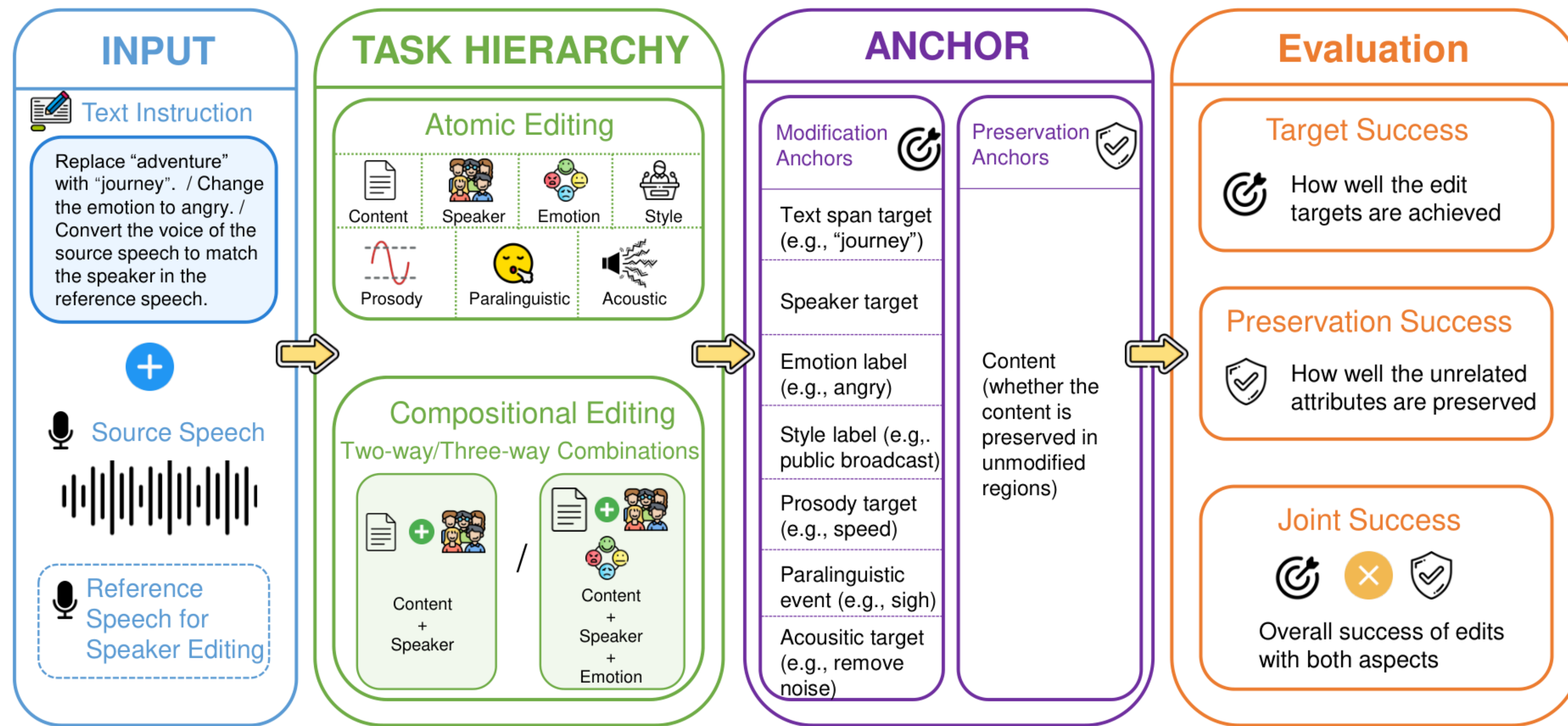
Balanced English and Mandarin data reveal model-dependent language bias.

3 Anchor-based metrics

Separate target success, preservation, and their conjunction.

Benchmark overview

Natural-language instruction + source speech → task-specific anchors → target, preservation, and joint success



Dataset at a glance

4,700

source-instruction pairs

4,300 + 400

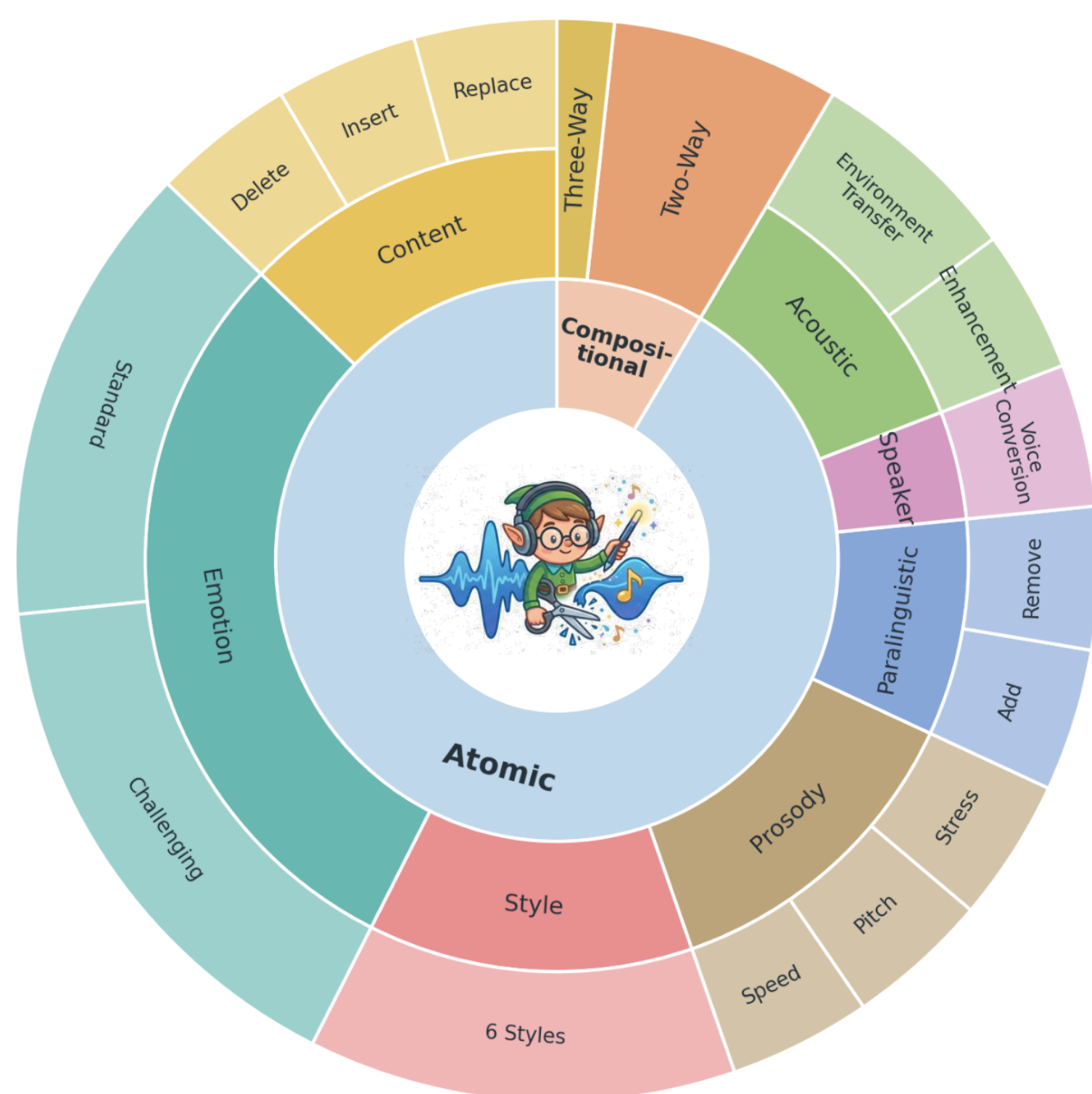
atomic + compositional

2,400 / 2,300

English / Mandarin

320 + 80

2-way / 3-way edits



Task distribution

Content

600

Speaker

200

Emotion

1,400

Style

600

Prosody

600

Paralinguistic

400

Acoustic

500

Compositional

400

Task → primary evaluator

Content target span → ASR alignment

Speaker reference voice → cosine ≥ .50

Emotion label → multimodal judge

Style label → target score ≥ 3

Prosody range → duration / F0 / prominence

Paralinguistic event → add / remove score

Acoustic condition → DNSMOS / RT60 / PANNs

Compositional components → all + joint success

Each sample uses task-specific anchors—not a single rigid reference waveform.

Anchor-based evaluation

t_i

Target success

p_i

Preservation success

j_i

Joint success

$$j_i = t_i \times p_i$$

Modification anchors: target span, speaker reference, emotion/style label, prosody range, event, acoustic condition.

Preservation gate: expected transcript WER/CER ≤ 10%.

Target vs. joint capability

Joint profiles contract when content preservation is enforced.

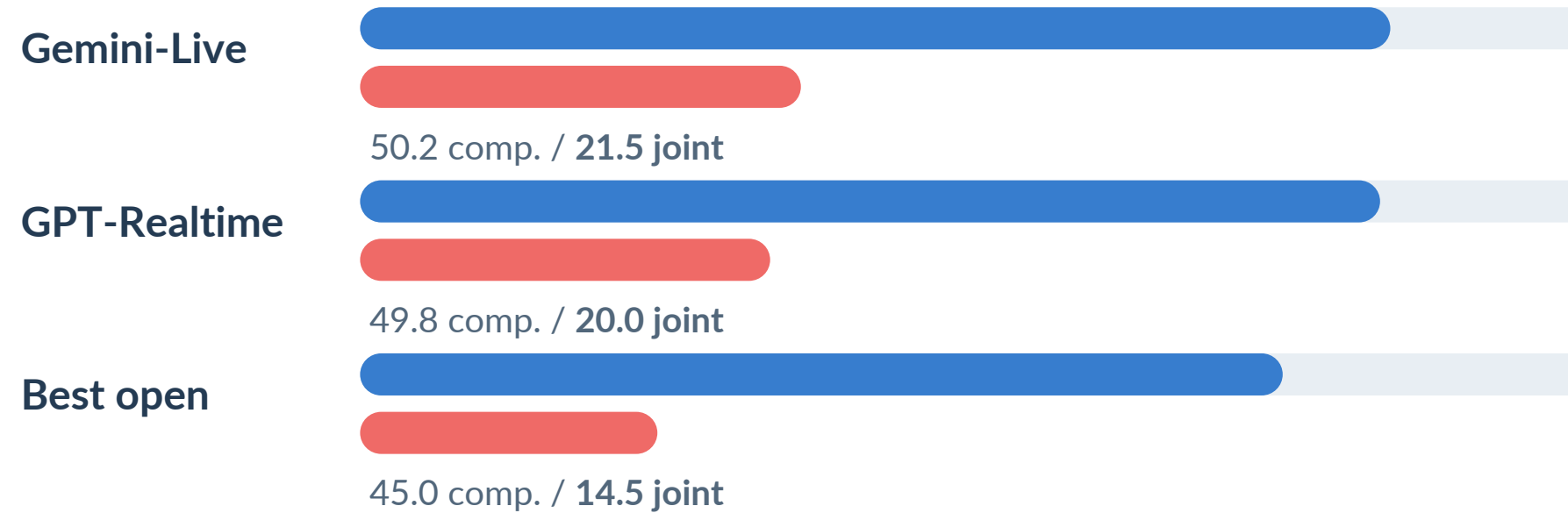


Compositional challenge

2-component, speaker-free subset

■ Component success

■ Joint success



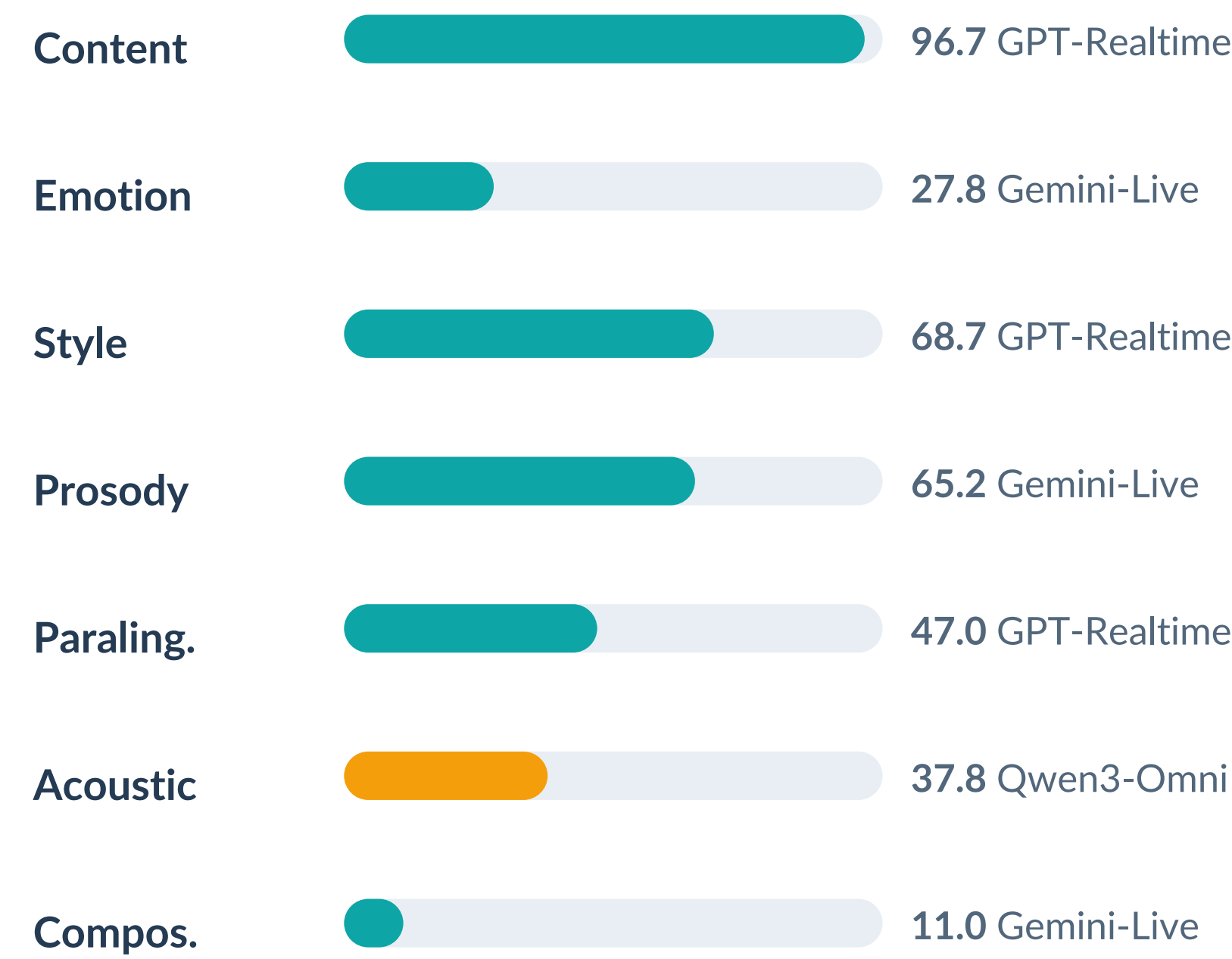
3-component: nearly every model reaches **0% joint success**.

Three findings

- 01 No universal winner — Strengths are task-specific.
- 02 Closed models lead — But Qwen3-Omni wins acoustic editing.
- 03 Preservation bottleneck — Target success can hide content corruption.

Best Speech LLM joint success

Best model varies by task → capability is fragmented.



Human validation

68.8%
agreement

240
outputs

720
judgments

Joint success · 95% CI 62.6–74.3
Pearson .685 · Spearman .699 (both p < .001)

Takeaway

Edit success alone is not enough.

Future speech editors must satisfy the instruction *and* preserve untargeted content—especially under multi-attribute commands.



SCAN FOR DATA + CODE

github.com/daxintan-cuhk/SpeechEditBench

arXiv:2606.01804 · Apache-2.0 code and benchmark assets

Questions? → tan.daxin1@huawei.com

SpeechEditBench v1.1 · 4,700 samples · English + Mandarin